Group 23 Game Proposal
Game Title: Graduation

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3. Favorite Game

Title: *Warcraft III* (Custom Tower Defense Maps)

Genre: Real-Time Strategy / Tower Defense

Description: *Warcraft III* custom maps pioneered the tower defense genre, emphasizing complex, maze-like enemy pathing and strategic tower placement. Our game draws heavy inspiration from these classic maps' focus on build variety and experimentation. Furthermore, just as *Warcraft III* utilizes powerful "Hero" units to anchor defenses, *Graduation* allows players to deploy "Professors" with unique skills and AoE strikes to synergize with their static towers.

4. General Information of the game

Graduation is at its core, a tower defense game with rogue-like elements with a huge emphasis on player build variety and experimentation between repeating playthroughs.
The story follows a normal year 4 student at HKU, as their graduation date looms ever closer, the school's SIS system goes rogue.

5. Game rules

The goal: Players must survive waves of "course monsters" across different HKU buildings to earn 6 credits per level, aiming to reach exactly 240 credits to graduate.
Difficulty tracks: Players select an academic track (Bachelor, Master, Doctorate) which scales enemy health, resource availability, and wave intensity.Enemies at higher difficulty levels may also have more complex gimmicks.
Professor Mechanics: Players can recruit and deploy Professors directly onto towers. These units provide localized passive buffs or grant the player active, cooldown-based abilities like massive AoE strikes or time-stopping fields.
Dynamic Maps: Completing all 5 levels of a specific faculty building grants a permanent, unique buff. Random encounters trigger unpredictably between waves, forcing players to adapt their builds on the fly.

6. The Four Basic Elements

Technology: Developed as a mobile game to ensure accessible controls for active skill deployment.
Story: A satirical narrative transforming universal university struggles into a physical battlefield.
Mechanics: Blends strategic tower placement with the high-stakes resource management of roguelikes and the active input of tactical RPGs via the Professor skills.
Aesthetics: A pixelated 2D art style that visually represents the "glitched" nature of the rogue SIS while keeping asset generation manageable.

7. Methodologies

The team will adopt an Agile methodology, focusing on rapid and iterative prototyping. We will prioritize developing a minimum viable product featuring one basic tower, one Professor ability, and one enemy type to establish the core game loop before expanding the map and enemy variety.

8. Platform and Tools

Game Engine: Unity
Programming Language: C#
Art & Assets:
Version Control: Github

Cloud Saves: Integration of Unity Gaming Services (UGS) or Firebase to support cross-platform cloud saving, allowing players to securely sync their academic progress (credit accumulation) across multiple devices

9. Tasks, Timeline and DoL

Week 1: Core Prototyping (Minimum Viable Product**)**
Establish the Unity project and GitHub repository.
Implement basic grid movement, placeholder tower placement, and simple enemy pathing.

Week 2: Mechanics & Abilities
Integrate the credit system (6 credits per level).
Program the "Professor" active skill mechanics and begin drafting pixel art assets for the first two HKU buildings.

Week 3: Level Design & Advanced Systems
Import custom pixel art into the engine and build out the first 10 playable floors.
Implement the dynamic building buffs and integrate the cross-platform cloud save backend.

Week 4: Random Encounters & Integration
Code and implement the Random Encounter events.
Connect all UI elements (menus, academic track selection, graduation screen).

Week 5: Playtesting & Final Polish
Conduct full-loop playtesting to balance enemy health pools and resource generation.
Fix bugs, polish audio/visual feedback, and finalize the submission build.

Wong Hei Nam Preston (Lead Programmer): Core Unity engine architecture, enemy pathing logic, and implementing the fundamental tower defense mechanics (placement, targeting, and shooting).
Tang Ho Lam (Game Designer & Systems Balancer): Designing the synergy system, detailing the "Professor" abilities (AoE strikes, time-stops), and balancing the Bachelor, Master, and Doctorate difficulty scaling.
Wu Kwun To Davis (Level Designer & Pixel Artist): Creating the pixel art assets (towers, monsters, UI elements) and mapping out the specific HKU building layouts (Engineering, Arts, Medicine, etc.) into playable grid maps.
Wong Kai Ching Oscar (Backend & Features Programmer): Setting up cross-platform cloud saves (via UGS/Firebase), coding the UI framework, and programming the trigger conditions for the rogue-like Random Encounters.